Data Attribution and Sources

Primary Data

• V-Dem (Varieties of Democracy) Country-Year Full+Others, version 15 (2024 data release). Indicators used: v2cltrnslw, v2juhcind, v2juncind, v2clkill, v2cltort, v2xcl_acjst, v2x_freexp, v2x_clphy, v2clrspct.

• World Bank World Development Indicators (WDI) via API — GDP per capita (current US$), population, prevalence of undernourishment, and poverty headcount at $2.15 (2017 PPP). Values are the most recent published as of atlas production (labeled 2026 baseline).

• United Nations Member States list (193 sovereign states) for country scope and ISO3 codes.

Source Framework

The RTLDI equation, the nine RTLP indicators, and the nested causal interpretation are taken from:

Sid J.A. Hubbard, Causality and Attraction: A Continuum of Steady States (Version 3, May 2026), DOI: 10.5281/zenodo.19468550. The operational crosswalk and binarization thresholds used here are documented in docs/indicator_crosswalk.md of this project.

This PDF was generated programmatically from the open RTLDI ATLAS toolkit. No proprietary or restricted data were used.

Nested Causal Modeling: Scope, Extensions, and Limits

The source document, Causality and Attraction: A Continuum of Steady States (Sid J.A. Hubbard, Version 3, May 2026, DOI 10.5281/zenodo.19468550), represents a serious, original attempt to give these protections a causal role inside a larger map of reality. Within its framework of nested causal enclosures and steady-state dynamic equilibria, the equal protection of the right to life is framed not merely as a correlate of prosperity but as a structural parameter of the "human enclosure" — an upstream condition whose strength or weakness shapes the higher-order economic and social steady states that can be reached and sustained.

This Atlas significantly extends that paradigm into the real world. We have taken the source equations and the nine RTLP indicators, operationalized them with transparent, reproducible data from V-Dem (2024 components for the eight governance and physical-integrity indicators) and the World Bank (2026-fresh GDP per capita baselines, population, and socioeconomic measures for the ninth indicator), and produced a complete diagnostic for every one of the 193 United Nations Member States. The addition of the nation-by-nation template, the contextual bounding study of archetypal cases (resource rentiers such as Qatar, late industrializers such as South Korea, global hubs such as Singapore, reformers such as Botswana and Rwanda, and high-R successes), the explicit 25% institutional-share cap on any single set of factors, the per-indicator attributable breakdowns, the three-year GDP-driven trends, the regional summaries with best/worst and universal-failure callouts, and the global and focus choropleths together constitute substantial new work in bridging the gap between the philosophical model and usable, globally comparable, data-rich diagnostics.

Nevertheless, this work does not — and cannot — provide direct proof of causation in the narrow policy-experiment sense. We cannot demonstrate that "doing X will cause Y to occur in Z time" at the nation-state or global level. There are no randomized controlled trials of national rights regimes at scale. The cross-sectional associations, the structural arguments drawn from the source, and even the carefully bounded empirical premium ($\eta = 0.30$ with the 25% cap) remain subject to omitted variables, reverse causality, selection, and the fundamental problem of constructing the counterfactual. The model supplies powerful benchmarks and a coherent interpretive lens; it does not deliver identified causal effects that would underwrite precise, time-bound predictions of the form "if this country flips these three indicators, its GDP will rise by $A billion by year N."

What the Atlas does provide is something more than bare correlation. It is a Nested Causal Modeling of an enormous amount of high-quality, real-world data — V-Dem expert-coded indicators cross-walked to the exact nine protections specified in the source, paired with the freshest economic baselines under the 2026 rule — all interpreted through the source's own architecture of nested causal enclosures. It goes as close to the real-world execution of causality as is currently possible for the simulations that must run in the minds of the people this atlas is for: citizens, civil society organizations, policymakers, finance ministries, media, and international partners who need to reason about costs, priorities, reform leverage, and the visible price of inaction.

We cannot furnish direct proof of the efficacy of large-scale institutional changes. The ones who will generate that proof — through the observable record of successes and failures in growth, resilience, investment, human development, and the avoidance of entropic costs — will be the nations that take the RTLDI seriously as a diagnostic tool and the nations that do not. The growth and prosperity of our states, and the hard, recurring costs of failing to protect human lives equally, are ultimately the direct responsibility of their people and the institutions they choose to sustain or reform.

It is not a new discovery that protecting human lives supports greater prosperity and steadier development. That relationship has long been observed across history and across literatures. What this Atlas makes newly legible, at global scale and with attributable component detail, is that refusing to protect human lives equally has a calculable, hard, annual cost — visible in the data, traceable to specific, fixable failures of the nine indicators, and bounded by realistic assessments of what any one family of institutional factors can explain once industry, resources, history, location, and human capital are given their due.

However far the RTLDI's accuracy may fall from perfectly right, it is not wrong. We have produced a new, transparent, globally applicable metric — grounded in the source paradigm, extended by the best available data, subjected to explicit contextual bounding, and offered as a tool for clearer thinking and more precise advocacy rather than as a finished causal proof. The ultimate test of its usefulness will be written not in these pages but in the real-world outcomes of the nations that choose to use it.

This appendix is offered in the same spirit as the source document: to make the structural costs of unequal protection of life more visible, more attributable, and more difficult to ignore.

Falsification of Malthusian Scarcity: Geodesic Populations and Equal Protection of Life

Thomas Malthus argued that human population tends to grow exponentially while food and other resources grow only linearly. The unavoidable result, in his view, would be recurring famine, poverty, and population checks through misery, vice, or war — unless population growth were deliberately limited.

The source document, Causality and Attraction (Hubbard 2026V3), offers a direct geometric and empirical counter rooted in the same nested causal framework that supplies the RTLDI. It draws on R. Buckminster Fuller's geodesic and tensegrity principles to treat human populations as collections of nodes within larger enclosing structures. In geodesic systems, as the frequency of subdivision (f) increases — that is, as more nodes and triangulated connections are added while maintaining consistent structural integrity — surface area scales as f^2 but load-bearing strength and overall system capability scale as f^3 or better. The addition of elements does not dilute the structure; under uniform connection rules, the whole becomes disproportionately stronger and more resilient.

The conditions Malthus projected — that greater population equates only to a greater drain on resources — are true but misleading, as they are not the whole truth. If nations do not equally protect the lives of all of the people within their borders, the population does behave in this way: more people equate to less available resource. But when the right to life is equally protected, GDP rises for a number of reasons and it rises exponentially via the f^3 scaling seen in geodesics. This is because societal stresses are distributed evenly among a more productive and more capable society.

The simple levers such as freedom from torture and freedom from arbitrary detention and an independent judiciary work together to make a nation that is investible — one that can attract businesses and stimulate domestic businesses to start up in an environment where the risk of death or arbitrary imprisonment are not stopping money from coming in. Similarly, whistleblower protections, once in place, would allow whistleblowers to come forward exposing corruption and theft at scales unknown before the protection was there.

The important distinction is that in falsifying Malthus in this way we have not negated the efficacy of Malthusian regimes and regulations. We have made it clear that a nation can choose the Malthusian relationship to their population and they can receive the expected drain on resources and plan for the expected disastrous trimming of the population to occur, or they could choose to put the protections in place to bring their RTLP score as high as it can and enjoy the f^3 cubic exponential scaling up of resilience, the conditions for capital investment to occur and for entrepreneurship to come from the people themselves as they would be an optimized enclosure capable of generating the resources it needs to prosper under the protections of a highly organized and stable society.

They can at least see the cost of not caring for the lives of their people equally and have a choice between the expansion of corruption or the stimulation of domestic industrial productivity. The linear drag term in the RTLDI equation quantifies the immediate cost of low frequency (missing protections). The geodesic perspective supplies the background for why raising that frequency can produce non-linear gains: it changes the scaling properties of the human system itself. This geometric and empirical counter-argument from the source, now expanded with the explicit choice framing, provides important context for the potential causal role of the RTLP indicators. The atlas measures the economic consequences of weak enclosures in the present data. The geodesic population model suggests why consistent, equal protection should enable the scaling behavior that turns added people into added capacity rather than added pressure. The ultimate test remains the real-world record of nations that maintain or improve equal protection versus those that do not.

This account draws directly from the source's treatment in the Malthus appendix, release notes, and geometric sections (especially the f^3 frequency scaling law as homology for human enclosures) while incorporating the conditional nature of the Malthusian outcome and the explicit choice between paths. It is offered here as background for the structural conditions under which the RTLDI's measured associations may reflect deeper causal architecture.

Index of Terms

2026-fresh G0:
: The most recent published GDP per capita values used as the dynamic baseline for the 2026 edition (even when RTLP components are from 2024).

Access to Justice:
: Fair access to remedies for right-to-life violations

Arbitrary Detention:
: Core protection against state overreach (RTLP indicator #4); failure contributes directly to attributable lost GDP in regions with weak rule of law.

Attributable lost GDP:
: The share of a country or region's total lost output that is directly traceable to the absence of one specific RTLP indicator (sum of $\eta/9 \times G_0 \times$ pop over countries failing that indicator, with current $\eta \approx 0.30$ from population-weighted cross-section).

Binarization / threshold:
: The documented conversion of continuous V-Dem and World Bank values into the binary 0/1 scores for the nine indicators (e.g., ≥2.0 on 0-4 scales, ≤5 % undernourishment + ≤10 % poverty for #9).

Causality and Attraction:
: The source monograph (Hubbard 2026V3, Zenodo 19468550) that supplies the RTLDI equation, the nine RTLP indicators, the nested-enclosures paradigm, and the cartographic critique of distortion.

Choropleth:
: Thematic map in which countries or regions are shaded by enclosure strength R (0–1) using the Viridis colormap; primary visual for the global, regional, and focus views.

Civilian Protection in Conflict Zones:
: Mechanisms protecting civilians in conflict areas

Crosswalk:
: The explicit mapping from the nine verbatim RTLP questions to observable V-Dem variables and World Bank statistics, with defined binarization thresholds.

ΔG (delta G):
: Annual GDP per capita loss due to incomplete right-to-life protection. Core output of the RTLDI equation.

Diagnostic (tool/guide):
: The use of RTLP scores, component breakdowns, and attributable lost-GDP figures as a practical instrument for citizens, NGOs, and policymakers to identify reform priorities and quantify upside.

Enclosure (nested causal / human):
: The source document's central metaphor: the interdependent, layered structures of protection whose strength or weakness shapes higher-order economic and social steady states.

η (eta):
: The sensitivity coefficient (current data-driven default 0.30 from population-weighted 2026 UN cross-section). In the bounded model $\Delta G = \min(\eta(1 - R) \times G_0, 0.25 \times G_0)$; the cross-sectional premium per indicator, then capped so the nine RTLP indicators are never credited with more than 25% of observed G_0 after other determinants are given due weight. (Source document used a more conservative 0.05 structural value.)

Existence of Legal Protections:
: Provisions explicitly protecting the right to life

Fitbounds / regional zoom:
: The per-nation choropleth view that automatically zooms to the target country plus the rest of its UN region, using the same Mollweide projection and Viridis scale as the global map.

Freedom from Torture and Inhumane Treatment:
: Measures in place to prevent torture

Freedom of Expression and Whistleblower Protections:
: Expression and whistleblower rights upheld

Frequency scaling (human enclosures):
: The application of geodesic f^3 principles to human systems: equal application of the nine RTLP protections raises the effective 'frequency' of the population structure, enabling non-linear gains in capability as population grows — directly challenging Malthusian predictions when enclosures are strong.

G_0 (G-zero):
: Baseline GDP per capita (World Bank, current US$). The 'current' economic size against which the deficit is measured.

Geodesic scaling / f^3 law:
: From Buckminster Fuller: in geodesic and tensegrity structures, as subdivision frequency (f) increases, surface area scales as f^2 while structural strength and capability scale as f^3 or higher. Used in the source as a homology for human populations: under equal protection of the right to life (consistent enclosure frequency), added people (nodes) increase systemic resilience and capacity disproportionately rather than triggering scarcity.

Independent Judiciary:
: Independent judiciary capable of upholding right-to-life laws

Law Enforcement Accountability:
: Law enforcement agencies accountable for unlawful killings

Malthusian theory:
: The proposition (Thomas Malthus) that population grows exponentially while resources grow linearly, inevitably producing scarcity and misery unless population is checked. The source and this atlas present geometric and empirical counter-evidence under conditions of equal right-to-life protection.

Member-nations table:
: The compact table on each regional summary page that lists every country in the region together with its R, G0, population, and total lost GDP, plus a bottom row of regional totals.

Mollweide:
: Equal-area pseudocylindrical map projection used for all choropleths (global, regional, and per-nation zooms) to ensure visual area corresponds to geographic reality.

Protection Against Arbitrary Detention:
: Arbitrary detention prohibited with legal recourse

R (RTLP score):
: Right-to-Life Protection score, 0–1. Average of nine binary indicators. 1 = full protection across all dimensions.

Right to Life:
: The overarching protected interest; the nine indicators operationalize equal and effective protection against arbitrary interruption.

RTLDI:
: Right-to-Life Deficit Index — the full framework and the per-country or global aggregate loss figure.

RTLP:
: Right-to-Life Protection — the nine binary indicators that are averaged to produce R.

Socioeconomic Conditions:
: Access to healthcare, food, and shelter adequate to sustain life

Steady state:
: A balanced biological, economic, or social condition that the framework argues is supported or undermined by the strength of the nine life protections and the resulting causal enclosures.

Torture and Inhumane Treatment:
: RTLP indicator #5; one of the largest contributors to global lost GDP, highlighting physical integrity failures.

Trend (3-year):
: The per-nation GDP-driven RTLDI plots (R held fixed at 2024 values) that illustrate recent economic-drag dynamics alongside the regional and global maps.

UN region / regional summary:
: One of the 22 geographic groupings used for aggregates, focused choropleths, member-nations tables (with R / G0 / population / lost + REGIONAL TOTAL row), and 9-indicator breakdowns.

V-Dem:
: Varieties of Democracy project. Supplies the expert-coded governance and civil liberties indicators for eight of the nine RTLP components.

Viridis:
: The perceptually uniform sequential colormap (dark low-R → yellow high-R) used consistently for every enclosure-strength choropleth in the atlas.

Credits and Acknowledgments

Conceptual Framework and Equations

Sid J.A. Hubbard — author of Causality and Attraction: A Continuum of Steady States (v3, 2026). All rights to the RTLDI equations, the nine RTLP indicators, and the nested causal paradigm remain with the author.

Data

• V-Dem Institute (https://www.v-dem.net) — V-Dem Country-Year dataset v15.
• World Bank Open Data (https://data.worldbank.org) — WDI via API.
• United Nations — list of 193 Member States.

Code and Atlas Production

RTLDI ATLAS project (2026). The open-source toolkit, crosswalk implementation, data pipeline, and this PDF generator are provided to make the framework usable by NGOs and researchers.

This work is released in the spirit of the original document: to map, with transparent data and reproducible math, the economic consequences of failing to protect the right to life equally.

For the full source code, 2026 data tables, and previous editions (2023, 2024), see the project repository. Users are encouraged to download the latest V-Dem and World Bank releases and re-run the pipeline for updated figures.